

Lab-2

Part I Culture media

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Culture Media used in Microbiology



Cultivation of bacteria

- The macroscopic study of microorganisms requires :
 1. Techniques for **isolating cells** from **natural** sources
 2. **Growing** them in the laboratory on **culture media**

Microbial cultivation: is a method of multiplying the Microorganisms in predetermined culture media under controlled lab. conditions

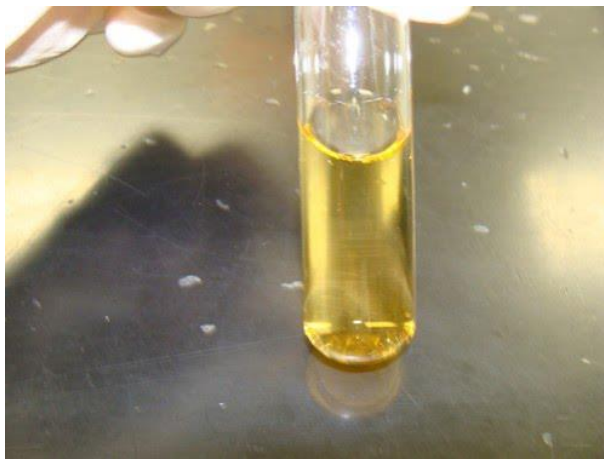
Why cultivate bacteria?

1. **Isolation.**
2. **Identification and characterization.**
3. **Determine which bacteria is most likely causing infection.**
4. **Test antimicrobial susceptibility .**
5. **Bank strain for future use .**

Definition Of Culture Media

The mixture of various nutrients that is suitable for the growth of microorganisms.

- Used for recognition and identification of microorganisms.
- Bacteria like any living cell it needs organic and inorganic material for their live.
- The media are contained either in sterile test tubes, plates (Petri Dishes), flasks or screw capped bottles.
- The media sterilized by moist heat.



Sterile Petri dish

Sterile screw capped

Basic Ingredient For Culture Media

1. **Peptone**: source of protein.
2. **Meat extract**: source for N₂ , it obtained by: Lean meat +H₂O Over 24h ,cold meat extracts
3. **NaCl**: for isotonic environment
4. **H₂O**: for humidity
5. **Agar Agar**: It is an unbranched polysaccharide that provides no nutritional benefits but used for solidifying purposes, it is obtained from sea weeds, melting (point is 92-95C°, solidifying point 42-45C°, conc. (1.5-2%) Not digested by M.O.
6. **pH**: most pathogenic bacteria needs a PH of 7.2-7.4, this obtained by addition of NaOH or HCL

Uses of media :

- **Simple media:** this media contains only common ingredients, used for cultivation of non fastidious m.o., e.g. nutrient broth & agar.
- **Special purpose media :** these media contains common ingredients with addition of another material.

Types Of Culture Media

Can be Classified based on :

- **A- Chemical composition**
- **B- Laboratory use**
- **C- Consistency**

A- Culture media based on chemical composition

- 1. Synthetic or Chemically Defined:** Exact chemical compositions are known
- 2. Complex Media:** Cannot write formula for each ingredient

B- Culture media based on laboratory use or purposes

- 1. Supportive media (Basic media) e.g. Nutrient agar**
- 2. Enriched media ...e.g. Blood agar**
- 3. Enrichment media ...e.g. Tetrathionate broth**
- 4. Selective media ...e.g. MacConkey agar , S-S agar**
- 5. Differential media ...e.g. MacConkey agar**
- 6. Transport media ...e.g. Sea Salt media**

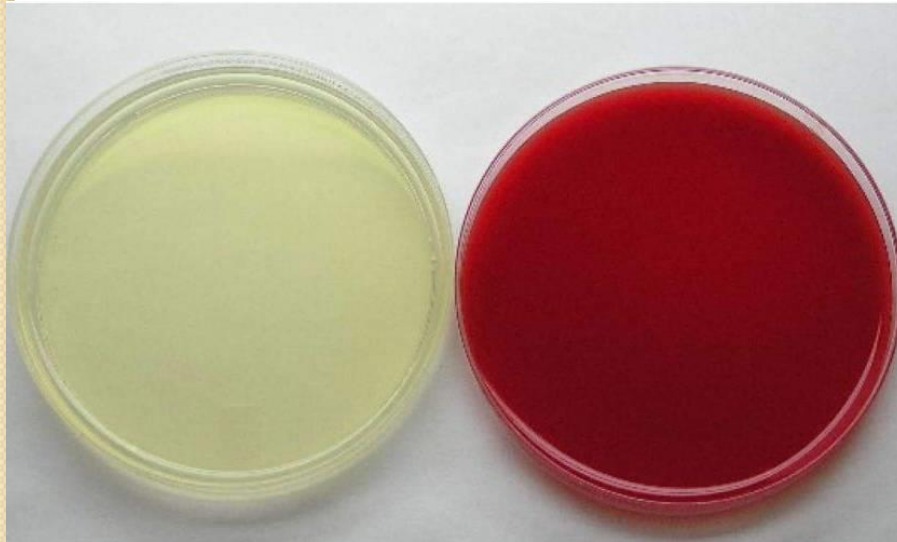
I - Supportive media (Basic media):

also known as general-purpose or basal media) are laboratory growth mediums that provide essential nutrients to support the growth of a wide variety of **non-fastidious** microorganisms without giving any special advantage to a specific group



2- Enriched medium: Simple media enriched with appropriate substances e.g., blood 5-10%, glucose 1-2%, serum and ascitic fluid 10% , are added to support some fastidious bacterial growth. e.g. blood agar.

- Most commonly used to cultivate **fastidious** m.o. like **Streptococci**.



Blood agar



Chocolate agar

- **Blood agar** is a general purpose enriched medium often **used to** grow fastidious organisms and to differentiate bacteria based on their **hemolytic properties**.



3- Selective media:

1. Containing inhibitory substance (e.g. bile salts, Antibiotic, dyes, ...etc)
 2. favor the growth of the certain microorganism and inhibit the growth of others.
- e.g. MacConkeys Agar, SS agar.



SS agar.



Cultured MacConkeys Agar

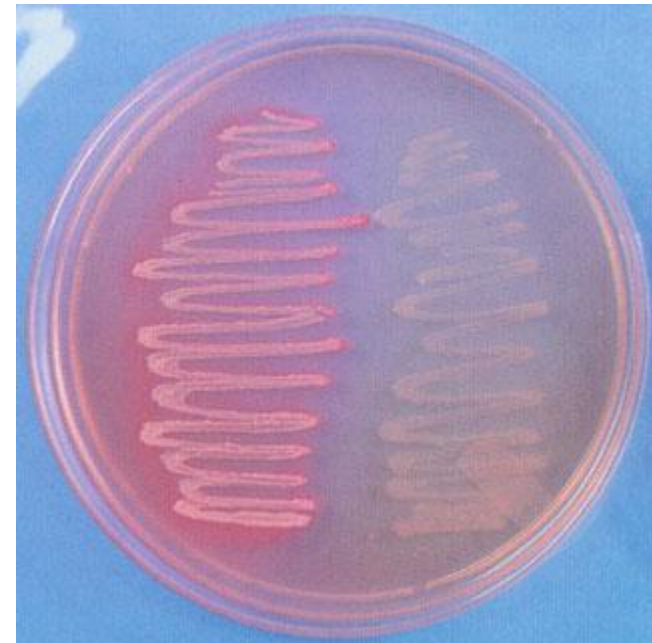
3 - Differential media

Certain bacterial species produce **characteristic growth** that can easily be recognized, or can produce certain **effects** in the medium

eg. Blood agar, MacConkeys Agar

Why differential?

Because it contains lactose which could differentiate between g+ve bacteria as two groups: lactose fermenter and non fermenter



MacConkeys Agar

4) Transport media

are essentially solutions of buffers with carbohydrates, peptones and other nutrients (excluding growth factors) designed to preserve the viability of **bacteria** during **transport** ... Depending on the type of organisms suspected in the sample.

- Special media for transport.
- e.g. Stuarts media.

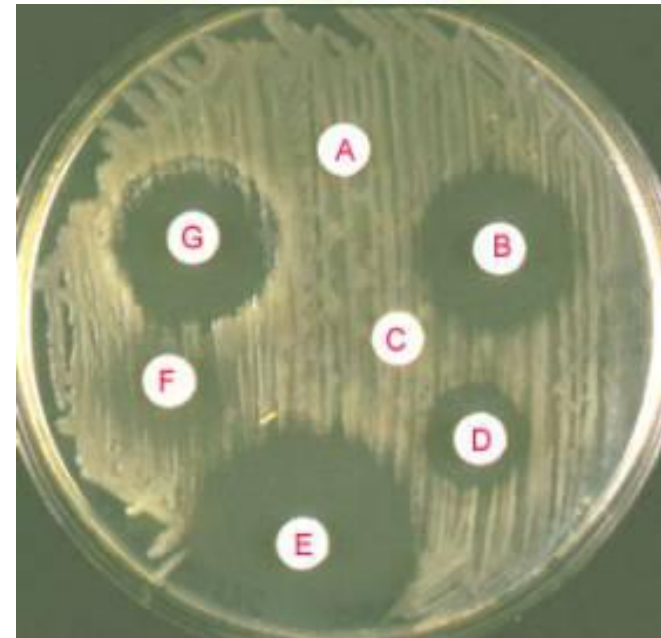
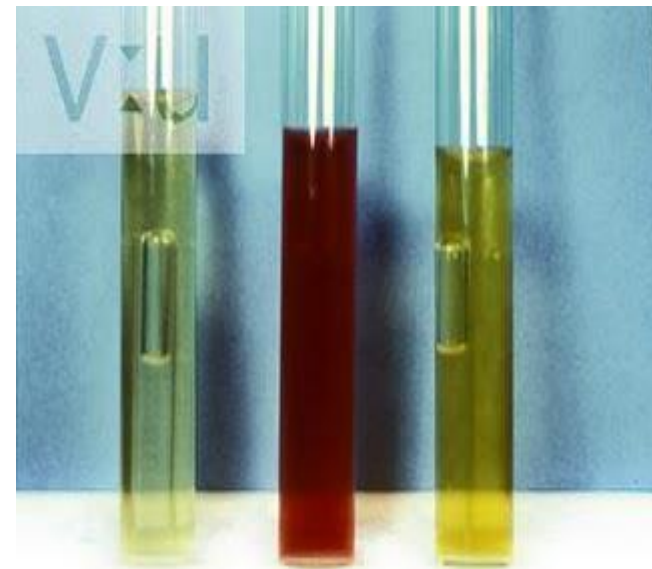


5- Indicator media

- This media use the **Visual** changes in the **color** of an **indicator** due to m.o. metabolism as a diagnosis feature.
- Like sugar media using Durhams` tube

6- Sensitivity media:

- A special media used to test antibiotic sensitivity.
- e.g. Muller hinton media.



Muller hinton media with antibiotic disc

C- Culture media based on consistency



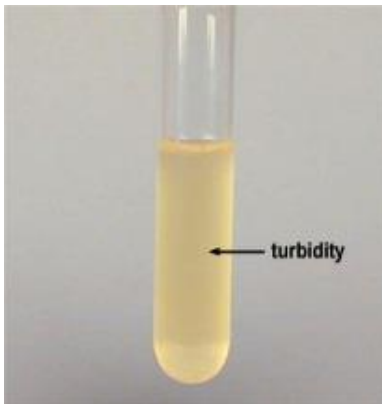
Liquid (broth) media

for primary cultivation

Disadvantage?

is the inability to obtain colony morphology

e.g., nutrient broth, peptone water, brain heart infusion

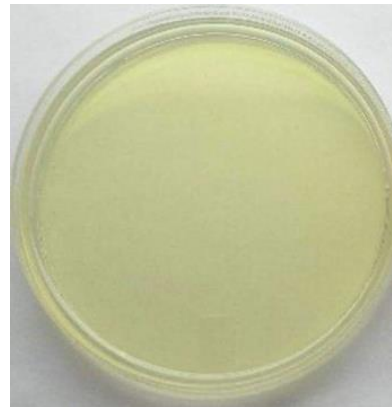


Solid media

for identification of colony morphology

conc. of agar is 1.5-2%

nutrient agar, MacConkeys agar, blood agar



Semisolid media

to study motility

Contain 0.4-0.8% of agar agar



**Thank
You**